

AI Model Development Phase 1

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Phase 1 Framing Brief: Small Language Models

Domain: Performance, Capabilities, and Comparative Advantages of Small Language Models (<7B parameters)

Main Research Question

What are the performance gaps and comparative advantages of small language models (<7B parameters) versus large language models, and what technical advances would enable small models to match or surpass larger models in specific domains?

Sub-Questions

1. **Performance Gaps:** Where do small models currently underperform large models, and by how much? (benchmarks, tasks, metrics)
2. **Current Advantages:** In which tasks or contexts do small models already match or exceed large models?
3. **Technical Enablers:** What techniques (distillation, pruning, efficient architectures) most effectively close the performance gap?
4. **Deployment Benefits:** What practical advantages (speed, cost, memory, privacy) give small models structural benefits?
5. **Fundamental Limits:** What do scaling laws and theoretical frameworks suggest about the limits of small vs. large models?
6. **Future Pathways:** What innovations could enable small models to surpass large models in reasoning and knowledge tasks?

Why This Question Matters

The Core Issue: If small models (<7B) can match large models (>100B) on most tasks, it would trigger a massive market disruption:

- **\$100B+ in LLM infrastructure becomes obsolete** - NVIDIA's H100 GPUs, massive data centers, OpenAI's model training investments
- **Power shifts from centralized AI labs to everyone** - Run GPT-4-level AI on your phone, no API fees, no data leaving your device
- **New winners emerge** - Companies optimizing small models (Mistral, Groq) beat API-dependent startups; edge chip makers (Qualcomm, Apple) gain over cloud providers (AWS, Azure)

This research answers: Are we headed for that disruption, or will large models maintain their moat?

What's at stake: Trillions in market cap, competitive positioning of every AI company, and whether AI becomes democratized or remains centralized.

Scope

IN SCOPE:

- Models <7B parameters: Phi-3, Gemma-7B, Llama-3-8B, Mistral-7B
- Comparison to large models: GPT-4, Claude-3, Llama-70B+ (>30B parameters)
- Performance benchmarks: MMLU, GSM8K, HumanEval, MATH, ARC, HellaSwag
- Technical methods: knowledge distillation, pruning, efficient architectures, mixture-of-experts
- Deployment metrics: inference speed, cost per token, memory footprint, latency
- Peer-reviewed papers (2023-2026), technical reports from major labs, benchmark studies

OUT OF SCOPE:

- Models >10B as "small" (clear boundary)
- Non-transformer architectures (unless hybrid with transformers)
- Pre-2023 models (focus on current state-of-the-art)
- Multilingual-specific studies (unless central to comparison)
- User satisfaction or UX research (maintain technical focus)